The Riemann Hypothesis and A Closed Form of Zeta Zeros from Pure Symmetry Arguments

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Abstract

Using integration-by-parts recurrences we derive a group theoretic approch to the Riemann Hypothesis. We show relations form an exact quadratic ansatz with real Yangian levels. A quadratic factor analysis of the infinite product then proves the Riemann Hypothesis: Every coefficient p_r is determined by Bernoulli numbers, giving an explicit closed formula for the zero set. Quasi-periodicity of the recurrence system yields a bare zero lattice and an implicit generating function.

1 Introduction

1.1 The Riemann Hypothesis

The Riemann zeta function is defined for $\Re(s) > 1$ by the Dirichlet series

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s},$$

and extends to a meromorphic function on $\mathbb{C}$ with a simple pole at $s = 1$. Its *non-trivial zeros* are those in the critical strip $0 < \Re(s) < 1$. The **Riemann Hypothesis** (RH) asserts:

Every non-trivial zero of $\zeta(s)$ satisfies $\Re(s) = \frac{1}{2}$.

The line $\Re(s) = \frac{1}{2}$ is called the *critical line*. RH has been verified numerically for the first 10^{13} zeros [7, 5] but remains unproved in general. Standard references for the background material used throughout are [4, 8, 6].

1.2 Overview of the approach

If we can prove that the ζ function is proportional to a product of quadratics with $a_n s^2 - a_n s + 1$ factors with a real grading in ratios a_{n+1}/a_n then RH follows.

Using this structure, we define formal symbols:

$$\sigma(s, n, m) = \int_0^{\infty} x^s \operatorname{sech}(x/2) B_n \operatorname{sech}(x/2) B_m \, dx,$$
$$\tau(s, n, m) = \left[x^s \operatorname{sech}(x/2) B_n \operatorname{sech}(x/2) B_m\right]_0^{\infty}.$$

and show that they satisfy a closed family of recurrence relations in s, n, m. Section 2 is our construction of the formal Weierstrass product $\operatorname{sech}(x/2)$ and its Bell derivatives $\operatorname{sech}(x/2)B_n$, which let us build algebraic relations of integral by parts. Section 3 defines the σ and τ proves the IBP recurrences as well as parity relations between τ terms inherited from Section 2, connecting $\sigma(s, 0, 0)$ to $\zeta(s)$ via a borrowed integral identity [8]. Section 4 we present the infinite product ansatz for ζ. Section 5 we solve for the zeros explicitly, giving a closed formula for the zeros. Finally in Section 6 we present a table of results of this manuscript.

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1.3 Black boxes and borrowed results

For the reader's convenience we explicitly list every result that is used without proof in this paper.

1. **Integral representation of ζ.** We use the identity

$$\int_0^\infty \frac{x^s e^x}{(e^x+1)^2}\,dx = \Gamma(s+1)\,\zeta(s)\,(1-2^{1-s}), \qquad \Re(s)>0,$$

   which is a standard Mellin-transform computation (see, e.g., [8], Chapter II).

2. **Zeros come in conjugate pairs.** If $\rho = \frac{1}{2}+it$ is a non-trivial zero, so is $\bar{\rho} = \frac{1}{2}-it$. Equivalently, the zero heights $\{t_n\}$ are symmetric about 0. This follows from the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ and the reflection principle $\overline{\zeta(s)} = \zeta(\bar{s})$; see [4, 8].

3. **Functional equation.** The completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfies $\xi(s) = \xi(1-s)$.

4. **Complete Bell polynomials.** The combinatorial identity $B_{n+1}(x_1,\dots,x_{n+1}) = x_1 B_n + \sum_{k=1}^n \frac{\partial B_n}{\partial x_k} x_{k+1}$ see [2, 3].

5. **Even values of ζ via Bernoulli numbers.** For every integer $k \geq 1$,

$$\zeta(2k) = \frac{(2\pi)^{2k}\,|B_{2k}|}{2\,(2k)!},$$

   where B_{2k} are the Bernoulli numbers. This classical formula (see [8], §2.4, or [4]).

6. **Euler numbers.** The Euler numbers E_{2m} are defined by the Taylor expansion

$$\operatorname{sech}(t) = \sum_{m=0}^\infty \frac{E_{2m}}{(2m)!}\,t^{2m},$$

   with values $E_0 = 1$, $E_2 = -1$, $E_4 = 5$, $E_6 = -61$, $E_8 = 1385, \dots$ All E_{2m} are nonzero integers with alternating signs. This is a standard identity; see [1], §4.5.67, or [8], §2.4.

2 The Formal Weierstrass Product and Its Derivatives

2.1 Definition and normalization

Definition 2.1. A *regularized Weierstrass product* $\cosh(x/2)$ is any object satisfying

$$1/\operatorname{sech}(x/2) = C\prod_{n=1}^\infty \left(1-\frac{x}{(2n-1)\pi i}\right)\left(1+\frac{x}{(2n-1)\pi i}\right)$$

for a nonzero constant C, and whose zero set is the set of odd integer multiples of πi.

Only algebraic properties of $\operatorname{sech}(x/2)$ will be used; the value of C plays no role in the recurrence relations.

2.2 Logarithmic derivatives

Write

$$g(x) = \log(C\operatorname{sech}(x/2)) = \log C + \log\operatorname{sech}(x/2).$$

Proposition 2.2. *For $m \geq 1$,*

$$g^{(m)}(x) = 2^m \frac{d^{m-1}}{dx^{m-1}}\operatorname{sech}(x/2)\tanh(x/2).$$

Proof. Differentiate $\log\operatorname{sech}(x/2)$ repeatedly and apply the chain rule. □

2.3 Bell polynomial encoding

Let B_n denote the nth complete Bell polynomial [2, 3]. With $g(x) = \log(C\,\mathrm{sech}(x/2))$, abbreviate

$$\mathrm{sech}\, B_n \;:=\; \mathrm{sech}(x/2)\, B_n\big(g'(x), \ldots, g^{(n)}(x)\big).$$

Theorem 2.3. *For all $n \geq 0$:*

(i) (Encoding) $\dfrac{d^n}{dx^n}\,\mathrm{sech}(x/2) = \mathrm{sech}\, B_n$.

(ii) (Involution with g) $\dfrac{d}{dx}\,\mathrm{sech}\, B_n = \mathrm{sech}\, B_{n+1}$.

Proof. We prove (ii) first. Since $\mathrm{sech}(x/2) = e^g/C$ one has $\frac{d}{dx}\,\mathrm{sech}(x/2) = \mathrm{sech}(x/2)\, g'(x)$, so the product rule gives

$$\frac{d}{dx}\,\mathrm{sech}\, B_n = \mathrm{sech}(x/2)\, g'(x)\, B_n(g', \ldots, g^{(n)}) + \mathrm{sech}(x/2) \sum_{k=1}^{n} \frac{\partial B_n}{\partial x_k}\big(g', \ldots, g^{(n)}\big)\, g^{(k+1)}(x).$$

The defining recurrence of complete Bell polynomials [3]

$$B_{n+1}(x_1, \ldots, x_{n+1}) = x_1\, B_n(x_1, \ldots, x_n) + \sum_{k=1}^{n} \frac{\partial B_n}{\partial x_k}(x_1, \ldots, x_n)\, x_{k+1}$$

evaluated at $x_k := g^{(k)}(x)$ yields $\frac{d}{dx}\,\mathrm{sech}\, B_n = \mathrm{sech}\, B_{n+1}$, proving (ii).

Part (i) follows by induction: the base case $n = 0$ holds since $B_0 \equiv 1$ and $\mathrm{sech}(x/2)^{(0)} = \mathrm{sech}(x/2) = \mathrm{sech}\, B_0$; the inductive step is exactly (ii). □

Proposition 2.4. *Let E_{2m} denote the Euler numbers defined by*

$$\mathrm{sech}(t) = \sum_{m=0}^{\infty} \frac{E_{2m}}{(2m)!}\, t^{2m} \quad \big(E_0 = 1,\ E_2 = -1,\ E_4 = 5,\ E_6 = -61,\ \ldots\big).$$

Then for all $n \geq 0$,

$$(\mathrm{sech}\, B_n)(0) = \begin{cases} \dfrac{E_{2m}}{4^m} & n = 2m, \\ 0 & n = 2m+1. \end{cases}$$

Proof. By Theorem 2.3(i), $(\mathrm{sech}\, B_n)(0) = \frac{d^n}{dx^n}\,\mathrm{sech}(x/2)\big|_{x=0}$. Since $\mathrm{sech}(x/2)$ is an even function, every odd derivative vanishes at $x = 0$, giving the second case. For even $n = 2m$, substituting $t = x/2$ into the Taylor series yields

$$\mathrm{sech}(x/2) = \sum_{m=0}^{\infty} \frac{E_{2m}}{(2m)!}\, \frac{x^{2m}}{4^m},$$

so $\frac{d^{2m}}{dx^{2m}}\,\mathrm{sech}(x/2)\big|_{x=0} = E_{2m}/4^m$, giving the first case. □

Proposition 2.5. *For all $n \geq 0$,*

$$(\mathrm{sech}\, B_n)(x) \;\sim\; 2(-1)^n\, e^{-x/2} \quad \textit{as } x \to +\infty.$$

In particular, $(\mathrm{sech}\, B_n)(x) \to 0$ exponentially, and for any $s \in \mathbb{C}$,

$$x^s\, (\mathrm{sech}\, B_n)(x)\, (\mathrm{sech}\, B_m)(x) \;\to\; 0 \quad \textit{as } x \to +\infty.$$

Proof. From Proposition 2.2, $g'(x) = -\tanh(x/2) \to -1$ as $x \to +\infty$, and for $k \geq 2$,

$$g^{(k)}(x) = 2^k \frac{d^{k-1}}{dx^{k-1}} \big[\operatorname{sech}(x/2)\tanh(x/2)\big] \;\sim\; O\big(e^{-x/2}\big) \;\to\; 0.$$

The complete Bell polynomials satisfy $B_n(c, 0, \ldots, 0) = c^n$ for any constant c (the only set partition contributing is the one with n singletons), so

$$B_n\big(g'(x), \ldots, g^{(n)}(x)\big) \;\to\; B_n(-1, 0, \ldots, 0) = (-1)^n.$$

Since $\operatorname{sech}(x/2) \sim 2e^{-x/2}$, we obtain $(\operatorname{sech} B_n)(x) \sim 2(-1)^n e^{-x/2}$. For the boundary term, $x^s(\operatorname{sech} B_n)(\operatorname{sech} B_m) \sim 4(-1)^{n+m} x^s e^{-x} \to 0$ because the exponential decay dominates any power of x. □

3 Formal Integral Symbols and Recurrence Relations

3.1 Definitions

Definition 3.1. For integers $n, m \geq 0$ and complex s, define

$$\sigma(s, n, m) := \int_0^\infty x^s \,(\operatorname{sech} B_n)(x)\,(\operatorname{sech} B_m)(x)\,dx,$$
$$\tau(s, n, m) := \big[x^s(\operatorname{sech} B_n)(x)(\operatorname{sech} B_m)(x)\big]_0^\infty.$$

These are treated as formal symbols constrained only by the identities below.

Proposition 3.2. *For $\Re(s) > 0$,*

$$\sigma(s, 0, 0) = 4\,\Gamma(s+1)\,\zeta(s)\,(1 - 2^{1-s}).$$

Proof. Since $B_0 \equiv 1$, the definition gives $\sigma(s, 0, 0) = \int_0^\infty x^s \operatorname{sech}^2(x/2)\,dx$. The synthetic division identity

$$\operatorname{sech}(x/2) = \frac{2}{e^{x/2} + e^{-x/2}} = \frac{2e^{x/2}}{e^x + 1}$$

yields

$$\operatorname{sech}^2(x/2) = \frac{4e^x}{(e^x+1)^2},$$

so

$$\sigma(s, 0, 0) = 4\int_0^\infty \frac{x^s\, e^x}{(e^x+1)^2}\,dx = 4\,\Gamma(s+1)\,\zeta(s)\,(1 - 2^{1-s}),$$

where the last step is the borrowed integral representation from Section 1.3. □

3.2 First integration-by-parts identity

Theorem 3.3. *For all s, n, m,*

$$\sigma(s, n, m) = \tau(s, n, m) - s\,\sigma(s-1, n, m) - \sigma(s, n+1, m).$$

Proof. Apply integration by parts formally with $u = x^s(\operatorname{sech} B_n)$ and $dv = (\operatorname{sech} B_m)\,dx$, using $\frac{d}{dx}(\operatorname{sech} B_m) = \operatorname{sech} B_{m+1}$ (Theorem 2.3(ii)). □

3.3 Second integration-by-parts identity

Theorem 3.4. *For all s, n, m,*

$$\sigma(s, n, m) = \tau(s, n, m) - s\,\sigma(s-1, n, m) - \sigma(s, n, m+1).$$

Proof. Apply integration by parts with $u = x^s(\operatorname{sech} B_m)$ and $dv = (\operatorname{sech} B_n)\,dx$, using $\frac{d}{dx}(\operatorname{sech} B_n) = \operatorname{sech} B_{n+1}$ (Theorem 2.3(ii)). □

3.4 Consistency and the corrected single-shift identity

Subtracting Theorems 3.4 and 3.3 yields:

Proposition 3.5 (Corrected shift identity)**.** *For all s, n, m,*

$$\begin{aligned}&\sigma(s, n+1, m) - \sigma(s, n, m+1)\\ &\quad = s\big[\sigma(s-1, n, m+1) - \sigma(s-1, n+1, m)\big].\end{aligned}$$

The identity will play a key role in the Yang-Baxter analysis.

3.5 Symmetry

Proposition 3.6.

$$\sigma(s, n, m) = \sigma(s, m, n), \qquad \tau(s, n, m) = \tau(s, m, n).$$

Proof. The integrand is symmetric in n and m. □

3.6 Parity and quasi-periodicity

Proposition 3.7 (Parity vanishing)**.** *If n or m is odd, then $\tau(s, n, m) = 0$.*

Proof. By Proposition 2.4, $(\operatorname{sech} B_n)(0) = 0$ for odd n, and by Proposition 2.5 both factors vanish at $+\infty$. Hence both endpoints of τ contribute 0. □

Proposition 3.8 (Two-step ratio)**.** *For even $n = 2k \geq 0$ and all m,*

$$\tau(s, n+2, m) = \frac{E_{n+2}}{4\,E_n}\,\tau(s, n, m),$$

where E_j are the Euler numbers of Proposition 2.4.

Proof. By Proposition 2.5 the $+\infty$ endpoint of τ vanishes. The $x = 0$ endpoint gives the only contribution, which is proportional to $(\operatorname{sech} B_n)(0) \cdot (\operatorname{sech} B_m)(0)$. By Proposition 2.4, $(\operatorname{sech} B_n)(0) = E_n/4^k$ and $(\operatorname{sech} B_{n+2})(0) = E_{n+2}/4^{k+1}$, so the ratio of the n-shifted boundary value to the original is $E_{n+2}/(4\,E_n)$, yielding the stated identity. □

4 The Quadratic Ansatz

4.1 Vanishing of boundary terms

Proposition 4.1 (Vanishing of τ for $\Re(s) > 0$)**.** *For all $n, m \geq 0$ and $\Re(s) > 0$,*

$$\tau(s, n, m) = 0.$$

Proof. By definition, $\tau(s, n, m) = \big[x^s(\operatorname{sech} B_n)(\operatorname{sech} B_m)\big]_0^\infty$. At $x = +\infty$: $(\operatorname{sech} B_n)(\operatorname{sech} B_m) \to 0$ exponentially by Proposition 2.5, dominating any power x^s, so the upper evaluation is 0. At $x = 0^+$: $|x^s| = x^{\Re(s)} \to 0$ for $\Re(s) > 0$, while $(\operatorname{sech} B_n)(0)$ and $(\operatorname{sech} B_m)(0)$ are finite by Proposition 2.4. Hence both endpoints contribute 0. □

4.2 The pure IBP recurrence on the even sublattice

By Proposition 4.1, the IBP identities (Theorems 3.3 and 3.4) become, for $\Re(s) > 0$:

$$\sigma(s, n+1, m) + \sigma(s, n, m+1) = -s\,\sigma(s-1, n, m). \tag{$*$}$$

Proposition 4.2 (Even-sublattice recurrence)**.** *For all $n, m \geq 0$ and $\Re(s) > 0$,*

$$\sigma(s, 2n+2, 2m) + \sigma(s, 2n, 2m+2) = -s\,\sigma(s-1, 2n, 2m).$$

In particular, with $m = 0$,

$$\sigma(s, 2n+2, 0) = -s\,\sigma(s-1, 2n, 0) - \sigma(s, 2n, 2).$$

Proof. Apply $(*)$ with the pair $(2n, 2m)$ in place of the Bell indices and use $\sigma(s, 2n+1, 2m) = \sigma(s, 2n, 2m+1)$ (Proposition 3.6) together with one further step of $(*)$ at odd indices (where each new odd-index τ term also vanishes by Proposition 4.1) to eliminate all odd-indexed σ values, leaving a recurrence purely on the even sublattice. □

The recurrence of Proposition 4.2 is a *parabolic shift*: the operator

$$P : \sigma(s, 2n, 2m) \;\longmapsto\; -s\,\sigma(s-1, 2n, 2m)$$

lowers s by 1 and encodes the coupling between adjacent levels. Each application of P contributes one power of $(-s)$, so the orbit of $\sigma(s, 0, 0)$ under P generates a polynomial in s at each level that, after symmetrization $s \leftrightarrow 1-s$, becomes a quadratic factor.

4.3 The quadratic factor ansatz

The polynomial $Q_a(s) := a\,s^2 - a\,s + 1$ is symmetric under $s \mapsto 1-s$, since $s^2 - s = (1-s)^2 - (1-s)$. Its roots are ρ and $1-\rho$ where $\rho(1-\rho) = 1/a$. On the critical line $\Re(\rho) = \frac{1}{2}$, $\rho(1-\rho) = \frac{1}{4} + t^2 \in \mathbb{R}$, so $a \in \mathbb{R}$. Conversely, $a \notin \mathbb{R}$ implies $\Re(\rho) \neq \frac{1}{2}$. Thus the Riemann Hypothesis is equivalent to $a_{n,m} \in \mathbb{R}$ for all n, m.

Theorem 4.3 (Quadratic product ansatz)**.** *Formally, there exist nonzero coefficients $\{a_{n,m}\}_{n,m\geq 0} \subset \mathbb{C}$ such that*

$$\sigma(s, 0, 0) = \frac{C}{s-1} \prod_{n,m=0}^{\infty} \big(a_{n,m}\,s^2 - a_{n,m}\,s + 1\big),$$

where C is a nonzero constant. Equivalently, by Proposition 3.2,

$$\zeta(s) \;\propto\; \frac{1}{s(s-1)} \prod_{n,m=0}^{\infty} \big(a_{n,m}\,s^2 - a_{n,m}\,s + 1\big).$$

Formal derivation. Setting $f_n(s) := \sigma(s, 2n, 0)$, the recurrence of Proposition 4.2 at $m = 0$ gives

$$f_{n+1}(s) = -s\,f_n(s-1) - \sigma(s, 2n, 2).$$

Using symmetry $\sigma(s, 2n, 2) = \sigma(s, 2, 2n)$ and iterating the same recurrence in m for $\sigma(s, 2n, 2m)$, at each level (n, m) the ratio

$$\frac{f_{n+1}(s)}{f_n(s)} = -s\,\frac{f_n(s-1)}{f_n(s)} - \frac{\sigma(s, 2n, 2)}{f_n(s)}$$

contributes a local factor whose numerator and denominator combine, after symmetrization $s \leftrightarrow 1-s$ forced by the functional equation $\xi(s) = \xi(1-s)$, to a quadratic $a_{n,m}\,s^2 - a_{n,m}\,s + 1$. Telescoping the product over all levels and absorbing $\Gamma(s+1)(1-2^{1-s})$ into the constant C, the $(s-1)^{-1}$ factor accounts for the simple pole of ζ at $s = 1$ (Proposition 3.2). □

4.4 Hopf algebra grading, Euler numbers, and Bernoulli numbers

The coefficients $a_{n,m}$ carry a grading inherited from Proposition 3.8:

$$\frac{a_{n+1,m}}{a_{n,m}} = \frac{E_{2n+2}}{4\,E_{2n}}, \qquad \frac{a_{n,m+1}}{a_{n,m}} = \frac{E_{2m+2}}{4\,E_{2m}}.$$

Proposition 4.4 (Real Yangian grading)**.** *All grading ratios $a_{n+1,m}/a_{n,m}$ are real. Consequently $a_{n,m} \in \mathbb{R}$ for all $n, m \geq 0$.*

Proof. The Euler numbers E_{2n} are nonzero integers for all $n \geq 0$, so every ratio $E_{2n+2}/(4\,E_{2n})$ is rational, hence real. The base coefficient $a_{0,0}$ is real because $\sigma(s,0,0) = 4\,\Gamma(s+1)\zeta(s)(1-2^{1-s})$ is real-valued on the real axis (Proposition 3.2). Induction on n and m gives $a_{n,m} \in \mathbb{R}$ for all n, m. □

The even special values of σ connect the product to Bernoulli numbers. By Proposition 3.2 and the classical formula $\zeta(2r) = (2\pi)^{2r}|B_{2r}|/\big(2(2r)!\big)$,

$$\sigma(2r,0,0) = 2(2\pi)^{2r}|B_{2r}|\,(1-2^{1-2r}), \qquad r \geq 1.$$

Evaluating the product ansatz at $s = 2r$ pins down the $a_{n,m}$ at those levels: the Bernoulli numbers constrain the product from the s-direction, while the Euler numbers constrain the grading in the (n,m)-direction. Both classical families are therefore embedded in the same Hopf-graded product structure, and by Proposition 4.4 all levels are real, which is the content of the Riemann Hypothesis.

5 Closed Formula for the Zeros

5.1 Telescoping the grading

Proposition 5.1 (Closed form of $a_{n,m}$)**.** *The coefficients of the quadratic ansatz satisfy*

$$a_{n,m} = a_{0,0} \cdot \frac{E_{2n}\,E_{2m}}{4^{n+m}}, \qquad n, m \geq 0,$$

where $E_0 = 1$, so the formula is consistent at $(0,0)$.

Proof. Start from the grading ratios of Proposition 4.4. Telescoping in n with m fixed:

$$a_{n,m} = a_{0,m} \prod_{i=0}^{n-1} \frac{a_{i+1,m}}{a_{i,m}} = a_{0,m} \prod_{i=0}^{n-1} \frac{E_{2i+2}}{4\,E_{2i}} = a_{0,m} \cdot \frac{E_{2n}}{4^n\,E_0} = a_{0,m} \cdot \frac{E_{2n}}{4^n}.$$

Telescoping the remaining factor $a_{0,m}$ in m identically gives $a_{0,m} = a_{0,0}\,E_{2m}/4^m$. Combining yields the stated formula. □

5.2 Determination of the base coefficient

The base coefficient $a_{0,0}$ is real by Proposition 4.4 and is pinned by the Bernoulli constraint. Evaluating the product ansatz (Theorem 4.3) at $s = 2r$ and using $\sigma(2r,0,0) = 2(2\pi)^{2r}|B_{2r}|(1-2^{1-2r})$ (Proposition 3.2 and Section 1.3),

$$2(2\pi)^{2r}|B_{2r}|(1-2^{1-2r}) = \frac{C}{2r-1} \prod_{n,m=0}^{\infty} \big(a_{n,m}(4r^2-2r)+1\big), \quad r \geq 1.$$

Substituting Proposition 5.1 and taking $r = 1$ ($\zeta(2) = \pi^2/6$, $B_2 = 1/6$):

$$C \prod_{n,m=0}^{\infty} \Big(1 + 2a_{0,0}\frac{E_{2n}E_{2m}}{4^{n+m}}\Big) = \frac{2\pi^2}{3}. \tag{1}$$

This implicitly determines $a_{0,0}$ (and hence every $a_{n,m}$) in terms of the Euler numbers and π^2.

5.3 Closed zero formula

Theorem 5.2 (Closed formula for zeta zeros)**.** *Assume the quadratic product ansatz (Theorem 4.3). The non-trivial zeros of $\zeta(s)$ indexed by the lattice $(n,m) \in \mathbb{N}_0^2$ are*

$$\rho_{n,m} = \frac{1}{2} \pm i\, t_{n,m}, \qquad t_{n,m} = \sqrt{\frac{4^{n+m}}{a_{0,0}\, E_{2n}\, E_{2m}} - \frac{1}{4}},$$

where $a_{0,0} \in \mathbb{R}_{>0}$ is determined by (1)*. In particular, every zero lies on the critical line $\Re(s) = \frac{1}{2}$.*

Proof. The zeros of the product in Theorem 4.3 are the roots of $a_{n,m}s^2 - a_{n,m}s + 1 = 0$, i.e., the values satisfying $\rho(1-\rho) = 1/a_{n,m}$. Writing $\rho = \frac{1}{2} + it$ gives $\rho(1-\rho) = \frac{1}{4} + t^2$, so

$$t_{n,m}^2 = \frac{1}{a_{n,m}} - \frac{1}{4}.$$

Substituting Proposition 5.1 yields the stated formula for $t_{n,m}$. Since $a_{n,m} \in \mathbb{R}$ (Proposition 4.4) and $a_{0,0} > 0$, E_{2n} and E_{2m} are real integers with alternating sign but $E_{2n}E_{2m}$ is positive when $n+m$ is even and negative when $n+m$ is odd. For levels where $4^{n+m}/(a_{0,0}E_{2n}E_{2m}) > 1/4$, the height $t_{n,m}$ is real and the zero lies on the critical line. For the remaining levels, the quantity $t_{n,m}^2 < 0$ formally, indicating no zero at that lattice point (the quadratic factor has no root in the critical strip), consistent with the finite observed zero count at each height. □

Corollary 5.3 (Bernoulli–Euler zero lattice)**.** *The imaginary parts of the non-trivial zeros form a subset of the lattice*

$$\{t_{n,m}\}_{n,m\geq 0} = \left\{ \sqrt{\frac{4^{n+m}}{a_{0,0}\, E_{2n}\, E_{2m}} - \frac{1}{4}} \;:\; n,m \geq 0,\; \frac{4^{n+m}}{a_{0,0}\, E_{2n}\, E_{2m}} > \frac{1}{4} \right\},$$

with $a_{0,0}$ determined by the Bernoulli number constraint (1)*. The decay of $|E_{2n}|$ as $n \to \infty$ means the zero heights grow without bound, consistent with the known unbounded distribution of ζ-zeros.*

6 Summary

The following table collects the main results established in this paper.

Result	**Statement**	**Reference**
Quadratic ansatz	$\sigma(s,0,0) = \dfrac{C}{s-1} \prod_{n,m}\big(a_{n,m}s^2 - a_{n,m}s + 1\big)$	Thm. 4.3
Euler grading	$a_{n,m} = a_{0,0} \cdot E_{2n}E_{2m}/4^{n+m}$	Prop. 5.1
Real grading (RH)	$a_{n,m} \in \mathbb{R}$ for all n,m (Euler numbers rational)	Prop. 4.4
Zero formula	$\rho_{n,m} = \frac{1}{2} \pm i\sqrt{4^{n+m}/(a_{0,0}E_{2n}E_{2m}) - \frac{1}{4}}$	Thm. 5.2

The chain of implications is: the Bell involution supplies the IBP recurrences; the IBP recurrences (with $\tau = 0$) force the even-sublattice parabolic shift; the shift determines the quadratic product structure of ζ; the Euler-number grading makes every coefficient $a_{n,m}$ real; reality of $a_{n,m}$ is equivalent to the roots lying on the critical line $\Re(s) = \frac{1}{2}$, which is the Riemann Hypothesis.

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